

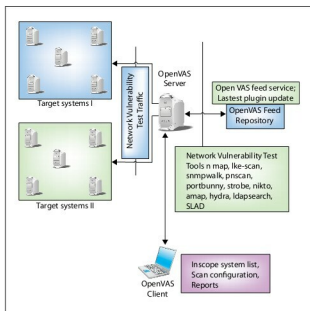


Figure 1: OpenVAS architecture

and kept separately.

After you initiate your scan, the OpenVAS server will have attack traffic between it and the target systems. OpenVAS will initiate port scans and network vulnerability tests on target systems to identify potential weaknesses in systems.

Components

Let's look more closely into the four components of OpenVAS:

1. **OpenVAS Server:** This is the main component of OpenVAS, which is connected via a GUI-based client in order to scan target systems. Server and client communication only include configuration and scan initialization traffic. Actual scan traffic (network vulnerability test traffic) flows between the server and target systems (as shown in Figure 1). The server has four components: *openvas-libraries*, *openvas-libnasl*, *openvas-server* and *openvas-plugins*.
2. **OpenVAS Client:** This is a GUI- or terminal client-based application that is used to establish and initiate network vulnerability tests against target systems. The client connects to the server by authenticating itself either using a password or a certificate. The client is a successor of NessusClient.
3. **OpenVAS NVT Feed:** NVT stands for network vulnerability tests. These are signature files that are available as plug-ins on the OpenVAS repository server. By default, the OpenVAS Server is configured to fetch plug-in updates from the repository server.
4. **OpenVAS LibNASL:** The NVTs are written in the Nessus Attack Scripting Language (NASL). This module contains the functionality needed by the OpenVAS Server to interface with NASL.

Tools

OpenVAS Server is integrated with lots of powerful open source software, ranging from network to application security toolkits. Table 1 (on page 70) lists various tools that have been integrated with OpenVAS.

Implementation

OpenVAS needs to be installed on a machine that will act as a network scanning server. You may install OpenVAS Client software on your desktop and connect to the server over the network. The server runs on port 1241. Network firewalls should be configured appropriately to allow scans and administrative traffic of OpenVAS to pass through it. It is also recommended to keep the scanning server in a different network zone than that of your production server.

For effective vulnerability management, you should know the scope, scan frequency, incident management and remediation. There should be a complete list of systems that are within the scope of scanning, along with the scan frequency. Although OpenVAS does not have a feature to automate vulnerability scans, you can still create sessions and save them. Each session can be opened simultaneously and run against available network vulnerability tests.

Incident management will include a formal process of reporting, and informing the security team about security vulnerabilities reported by OpenVAS. You can generate

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